# ERD of the Control Group Consortium

**LucidChart link: here**

## Institution and People

### institution
- **institutionId PK**
- institutionName
- institutionType

### department
- **departmentId PK**
- *institutionId FK*
- departmentName

### payer
- **payerId PK**
- *personId FK*
- *institutionId FK*
- *departmentId FK*

### affiliation
- **affiliationId PK**
- *personId FK*
- *institutionId FK*
- validFrom
- validTo

### person
- **personId PK**
- personName
- email
- phone

### role
- **roleId PK**
- roleName
- canViewConfidential

## Resources

### resource_type
- **resourceTypeId PK**
- resourceTypeName
- resourceTypeKind

### qualification
- **qualificationId PK**
- qualificationName
- qualificationKind

### person_qualification
- **personQualificationId PK**
- *personId FK*
- *qualificationId FK*
- validFrom
- validTo
- Issuer

### site
- **siteId PK**
- *institutionId FK*
- siteName
- siteTimeZone

### resource
- **resourceId PK**
- *resourceTypeId FK*
- *siteId FK*
- resourceName
- serialNumber
- movable
- capability
- limitation
- maxSessionMinutes

### resource_requirement
- ***resourceId PK FK***
- ***qualificationId PK FK***

### equipment_placement
- **equipmentPlacementId PK**
- *equipmentId FK*
- *facilityId FK*
- startAt
- endAt

### configuration
- **configurationId PK**
- *facilityId FK*
- configurationName
- configurationPurpose

## Access Control

### credential
- **credentialId PK**
- *personId FK*
- *issuerInstitutionId FK*
- externalSubject
- validFrom
- validTo

### access_grant
- **accessGrantId PK**
- *personId FK*
- *resourceId FK*
- *projectId FK*
- *authorizedBy FK*
- validFrom
- validTo

### person_role
- ***personId PK FK***
- ***roleId PK FK***

### access_log
- **accessLogId PK**
- *credentialId FK*
- *resourceId FK*
- *eventId FK*
- occurredAt
- result
- reason

## Project & Request

### project
- **projectId PK**
- *principalInvestigatorId FK*
- projectCode
- projectTitle
- projectVisibility

### project_member
- ***projectId PK FK***
- ***personId PK FK***
- responsibility

### experiment
- **experimentId PK**
- *projectId FK*
- *leadResearcherId FK*
- experimentName
- startAt
- endAt

### request
- **requestId PK**
- *experimentId FK*
- *requestingInstitutionId FK*
- *requesterId FK*
- submittedAt
- desiredStart
- desiredEnd
- staffingMinutes
- priority
- safetyNotes
- specialConditions
- expectedOutcome

### request_need
- ***requestId PK FK***
- ***resourceTypeId PK FK***
- quantity
- estimatedMinutes

### decision
- **decisionId PK**
- *requestId FK*
- *eventId FK*
- *decidedBy FK*
- decidedAt
- outcome
- rationale

## Event

### event
- **eventId PK**
- *experimentId FK*
- eventType
- startAt
- endAt
- eventStatus

### event_resource
- ***eventId PK FK***
- ***resourceId PK FK***

### event_configuration
- ***eventId PK FK***
- ***configurationId PK FK***

### staff_assignment
- ***eventId PK FK***
- ***personId PK FK***
- duty

### event_change
- **eventChangeId PK**
- *eventId FK*
- *changedBy FK*
- changedAt
- oldStart
- oldEnd
- oldStatus
- newStart
- newEnd
- newStatus
- reason

### staff_availability
- **staffAvailabilityId PK**
- *personId FK*
- startAt
- endAt

## Pricing and Allocation

### rate
- **rateId PK**
- *resourceTypeId FK*
- *consumableId FK*
- staffService
- unitPrice
- validFrom
- validTo

### estimate
- **estimateId PK**
- *eventId FK*
- *rateId FK*
- quantity
- estimationTime
- assumption

### consumable
- **consumableId PK**
- consumableName
- unit

### funding_eligibility
- ***fundingSourceId PK FK***
- ***rateId PK FK***

### funding_source
- **fundingSourceId PK**
- *payerId FK*
- fundingSourceName
- fundingSourceType
- budget
- validFrom
- validTo

### policy_share
- **policyShareId PK**
- *fundingSourceId FK*
- *allocationPolicyId FK*
- *beneficiaryProjectId FK*
- basisPoint

### allocation_policy
- **allocationPolicyId PK**
- *projectId FK*
- *authorizedBy FK*
- validFrom
- validTo
- rationale

## Billing

### usage_record
- **usageId PK**
- *eventId FK*
- *resourceId FK*
- *staffPersonId FK*
- *consumableId FK*
- quantity
- recordedAt

### charge
- **chargeId PK**
- *usageId FK*
- *rateId FK*
- *allocationPolicyId FK*
- state
- createdAt

### adjustment
- **adjustmentId PK**
- *chargeId FK*
- *authorizedBy FK*
- authorizedAt
- amount
- reason

### charge_allocation
- **chargeAllocationId PK**
- *chargeId FK*
- *policyShareId FK*
- amount

### invoice_lines
- ***chargeAllocationId PK FK***
- ***invoiceId PK FK***

### invoice
- **invoiceId PK**
- *fundingSourceId FK*
- periodStart
- periodEnd
- issuedAt
- status